



Advances in Non-Invasive Blood Glucose Monitoring: A Comprehensive Review

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Abstract

Despite extensive research into non-invasive blood glucose monitoring (NIBGM), existing reviews tend to address optical or microwave techniques separately, lacking an integrated analysis of their convergence into IoT-based wearable systems. This review fills that gap by systematically evaluating recent advances (2018–2025) in hybrid sensing technologies, with a particular focus on Red-Near Infrared (R-NIR) spectroscopy, microwave-based sensors, and their integration into real-time, wearable Internet-of-Things (IoT) platforms. Recent advancements in biosensor design, employing optical sensing within the 630–850 nm range combined with cloud-based analytics, have demonstrated clinical accuracy up to 98.82% compared to invasive standards ($R^2 = 0.96$). Microwave resonator-based sensors, particularly those utilizing complementary split ring resonators (CSRRs), have similarly emerged as promising non-invasive techniques, showing improved sensitivity through near-field configurations. By critically analyzing sensor architectures, calibration methods, dielectric modeling, and IoT integration strategies, this review highlights pathways toward scalable and clinically relevant NIBGM solutions. Ultimately, the paper provides an integrated perspective aimed at guiding future development of reliable, AI-enhanced, and patient-friendly glucose monitoring devices within emerging smart healthcare ecosystems. This review synthesizes **112** peer-reviewed studies (2018–2025) spanning optical (R–NIR, Raman, MIR/THz), microwave/metamaterial RF, hybrid sensing, and AI/IoT-integrated systems to provide an integrated perspective on clinically viable NIBGM pathways.

Keywords: Non-invasive blood glucose monitoring; Red-near infrared spectroscopy; Microwave sensing; Wearable biosensors; Internet-of-things integration.

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1. Introduction

According to the World Health Organization (WHO), diabetes affects approximately 450 million people world-wide, with over 150 million cases in China alone.^[1] Continuous blood glucose monitoring is critical for managing fluctuations, especially in brittle diabetes cases. Traditional electrochemical methods, such as Self-Monitoring Blood Glucose (SMBG) and Continuous Glucose Monitoring (CGM) sensors, remain invasive. Non-invasive alternatives utilizing optical and microwave technologies are being explored, but accuracy challenges remain. Future research focuses on improving sensor sensitivity, integrating AI, and enhancing real-time monitoring capabilities.^[1,2]

Despite these developments, challenges persist in achieving

the accuracy and reliability required for clinical application. Factors such as individual physiological differences, environmental conditions, and the complexity of glucose dynamics complicate the development of universally applicable non-invasive monitoring systems. Ongoing research, aims to address these issues by refining sensor technologies, improving data analysis methods, and integrating advanced algorithms to enhance predictive accuracy.^[2,3]

Recent advancements in non-invasive blood glucose monitoring have aimed to overcome the limitations of traditional invasive techniques. A high-figure of merit (FoM) plasmonic microwave resonator was introduced as an on-body non-invasive glucose sensor, operating at resonance frequencies of 3.25 GHz and 4.67 GHz, demonstrating a fivefold sensitivity improvement over conventional designs. Experimental tests conducted on human subjects validated the sensor's precision in monitoring interstitial fluid glucose variations, highlighting the potential of microwave-based sensors for diabetes management.^[4]

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Another approach involves a 3D-printed microwave sensor that enables non-invasive glucose detection in saliva. Made using Aerosol Jet Printing (AJP) on glass, it features an interdigital capacitor for high sensitivity. Operating in the K-band, it detects glucose levels as low as 10 mg per 100 ml, making it promising for real-time, wearable monitoring.^[5]

Microwave imaging techniques have also been explored for non-invasive glucose monitoring in mice. A study utilizing thermo-elastic optical indicator microscopy (TEOIM) visualized microwave near-field intensity variations around the mouse tail, correlating them with blood glucose levels. Experimental results confirmed a mean absolute relative difference (MARD) of 7.05% and a root mean squared error (RMSE) of 24.85 mg/dL, indicating promising continuous monitoring capabilities.^[6]

In another study, Near-Infrared Spectroscopy (NIR) was integrated with a MAX30102 biosensor and ESP32 microcontroller for non-invasive glucose monitoring. The system captured photoplethysmogram (PPG) signals, processed via linear regression with Clarke Error Grid Analysis. The device achieved a root mean squared error (RMSE) of 13.8 mg/dL and a mean absolute relative difference (MARD) of 12%, with real-time glucose data displayed on an OLED screen and uploaded via IoT.^[7]

A systematic review categorized non-invasive glucose monitoring techniques into four classes: optical spectroscopy, photoacoustic spectroscopy, electromagnetic sensing, and nanomaterial-based detection. Electromagnetic based sensors demonstrated promising accuracy, while optical methods required calibration to mitigate interference. The study emphasized integrating AI and nanotechnology for sensor precision and real-time monitoring improvements.^[8]

A review of non-invasive glucose sensing technologies analyzed 243 studies, with 124 focusing on clinical validation and technological advancements. Key challenges include accuracy, environmental interference, and device calibration. Machine learning-assisted spectroscopy and photonic sensors showed potential in enhancing reliability, with future research emphasizing clinical trials and commercial deployment.^[9]

Another review categorized saliva, skin, and breath-based glucose sensing methods, evaluating their accuracy. Reverse iontophoresis, spectroscopy, and microwave-based sensing emerged as viable non-invasive alternatives. However, challenges such as calibration needs, environmental interference, and regulatory approvals persist. IoT and AI-driven analytics were identified as key enablers for real-time diabetes management.^[10]

Microwave-based sensors for glucose monitoring leverage dielectric property variations of human tissues. Dielectric spectroscopy at microwave frequencies showed that increasing glucose levels decrease relative permittivity and conductivity. Metamaterial resonators and impedance spectroscopy-based RF/microwave sensors were evaluated, with future advancements aimed at improving sensor accuracy, minimizing environmental interference, and developing

wearable solutions.^[11]

Metamaterial-based microwave sensors have demonstrated a fivefold increase in phase sensitivity of (S21) for glucose detection. A study utilizing a metamaterial sensor with nine unit cells at 1.99 GHz reported an average sensitivity of 3.66° per mmol/L, suggesting significant potential for real-time, non-invasive glucose monitoring and clinical applications.^[12]

Microwave biomedical sensors for glucose monitoring face instability challenges due to variations in skin and fat tissue properties. A proposed sensor design integrating a matching layer significantly improved resonance frequency stability and sensitivity, reinforcing the feasibility of microwave sensors for continuous glucose monitoring.^[13]

A comparative review classified non-invasive glucose monitoring into optical, microwave, and electro-chemical techniques. Optical approaches such as near-infrared and Raman spectroscopy offer high precision but suffer from signal interference. Microwave sensors leverage dielectric property variations, providing deeper tissue penetration. Electrochemical methods, including reverse iontophoresis, extract glucose from interstitial fluid. Future developments aim to integrate AI, biosensors, and wearable technology for accurate, real-time glucose tracking.^[14]

A study introduced a cost-effective, portable 2.4 GHz microwave glucose sensor utilizing metamaterial cells and an RF-DC rectifier circuit. The sensor achieved $\pm 2\%$ detection error, operates independently without bulky VNAs, and measures blood glucose via fingertip placement, offering a compact ($12 \times 17 \times 8 \text{ cm}^3$) and affordable alternative.^[15]

An IoT-enabled real-time glucose monitoring framework (iGM) utilizing red and near-infrared (R-NIR) spectroscopy was also developed. Built on the ESP32 platform, the system streams glucose data to AWS IoT Core. Clinical results showed 98.82% accuracy after fasting and 98.04% accuracy post-meal, demonstrating IoT's role in continuous, non-invasive glucose monitoring for diabetes management.^[16]

Ahmed A. Zakaria *et al.*^[17] study investigates the influence of resonator proximity on the sensitivity of non-invasive microwave-based glucose sensors fabricated on Rogers 4360G2 substrates. Utilizing complementary split ring resonators (CSRRs), the research compares near and far resonator placements through S21 parameter analysis, revealing that closer resonator configurations exhibit significantly higher sensitivity to glucose concentration changes. The study supports the optimization of resonator design and placement for improved sensor efficiency, particularly in detecting subtle physiological variations critical for diabetes management.^[17]

In this paper,^[18,19] a planar microwave glucose sensor is proposed using dual CSRRs fabricated on a Rogers substrate to enhance non-invasive blood glucose detection. The authors examine different resonator placements (15 mm, 20 mm, and 25 mm) and identify 15 mm as the optimal distance, yielding the highest sensitivity. Simulations incorporating the Cole-

Cole model for biological tissues show that closer resonator spacing results in improved electromagnetic coupling and better responsiveness to glucose concentration variations, offering a promising advancement in wearable and pain-free glucose monitoring systems.

Despite the increasing number of research efforts in NIBGM, existing review articles often present fragmented perspectives—either focused on standalone sensor technologies or emphasizing clinical outcomes without covering sensor-level integration. Many reviews published prior to 2022 do not address the growing convergence between Red-Near Infrared (R-NIR) spectroscopy, microwave sensing, and IoT platforms. Additionally, the role of artificial intelligence in calibration and prediction has received limited systematic coverage.

This review fills that gap by offering a timely and comprehensive analysis of recent (2018–2025) innovations in NIBGM. It uniquely combines optical and electromagnetic sensing paradigms with discussions on real-time deployment in wearable and cloud-based environments. By bridging technical, clinical, and engineering perspectives, this review aims to guide researchers, device developers, and clinicians in developing next-generation smart diabetes management tools.

This review aims to provide a comprehensive, up-to-date synthesis of non-invasive blood-glucose monitoring (NIBGM) technologies, highlighting their sensing mechanisms, performance, and integration potential. Specifically, the objectives are to:

1. Compare the principles, advantages, and limitations of optical (R-NIR, Raman, MIR/THz), microwave/metamaterial, and hybrid multimodal approaches.
2. Examine the role of AI-driven calibration and IoT-based implementation in improving accuracy, real-time analysis, and data security.
3. Identify key research gaps and future directions, including multimodal integration and wearable design trends.

By articulating these aims, the review clarifies its analytical scope and distinguishes its contribution from earlier surveys that focused on a single sensing modality.

2. Review methodology

Scope update (April 2025): To ensure comprehensive coverage suitable for a state-of-the-art review, the database search was repeated in April 2025 and expanded to include IEEE Xplore, ScienceDirect, Springer-Link, PubMed, Web of Science, *Sensors* (MDPI), *Nature Biomedical Engineering*, *ACS Sensors*, and *Biosensors & Bioelectronics*. The final corpus comprises **112** peer-reviewed studies published between 2018 and 2025, spanning optical (R-NIR, Raman, MIR/THz), microwave/RF (including metamaterial CSRR and plasmonic designs), hybrid/multimodal systems, and AI/IoT-integrated implementations.

This review adopts a narrative-scoping hybrid approach to examine emerging technologies in non-invasive blood-glucose monitoring (NIBGM), emphasizing sensor

mechanisms, real-time integration, and clinical readiness.

Databases and search queries: Searches were executed on all selected databases using Boolean combinations of the following terms and their synonyms:

- “non-invasive glucose monitoring” AND (“microwave” OR “RF” OR “metamaterial” OR “CSRR” OR “dielectric”);
- (“R-NIR” OR “near-infrared” OR “Raman” OR “mid-infrared” OR “terahertz”) AND (“glucose” OR “diabetes”);
- (“photoplethysmography” OR “PPG”) AND (“machine learning” OR “deep learning” OR “PLSR” OR “transfer learning”);
- (“hybrid” OR “multimodal”) AND (“optical” AND “microwave”) AND (“sensor” OR “biosensor”);
- (“IoT” OR “wearable” OR “edge computing”) AND (“glucose monitoring” OR “CGM”).

Inclusion criteria:

- Published between January 2018 and March 2025;
- Experimental, clinical, or high-fidelity simulated NIBGM studies;
- Clear description of wearable, portable, or IoT-based implementation;
- Quantitative performance metrics reported (e.g. accuracy, RMSE, MARD, sensitivity, R^2);
- Explicit discussion of optical, microwave/RF, or hybrid sensing principles.

Exclusion criteria:

- Purely invasive or electrochemical methods;
- Patents, editorials, and non-validated concept papers;
- Non-English studies without official translation;
- Reports lacking sufficient methodological or quantitative detail.

Screening summary: After applying the above criteria, duplicates were removed and 112 publications were retained. [Table 1](#) summarizes the filtering process.

Categorization: Selected studies were grouped into four technological domains: (i) optical-based, (ii) microwave-based, (iii) hybrid multimodal, and (iv) AI- or IoT-integrated systems. Emphasis was placed on sensor architecture, detection principles, clinical performance, and integration feasibility for wearable healthcare applications.

Rationale for exclusion criteria: The exclusion criteria were defined to maintain methodological consistency and data reliability. Patents and editorials were omitted because they generally lack quantitative validation or reproducible methodologies. Purely invasive or electrochemical sensors were excluded to focus exclusively on non-invasive technologies. Non-English studies without verified translations were removed to avoid potential misinterpretation of technical terminology. Finally, studies without sufficient methodological or statistical detail were excluded to ensure comparability of reported performance metrics.

Compared with earlier reviews focusing exclusively on optical or electromagnetic domains, this extended corpus consolidates more than one hundred studies encompassing (i) AI-assisted calibration for physiological variability, (ii)

Table 1: Search and screening summary (PRISMA-style).

Steps	Records	Notes
Initial retrieval (2018–2025)	265	Multiple databases and journals
Duplicates removed	43	Title/abstract matching
Title/abstract screened	222	Inclusion/exclusion applied
Full-text assessed	138	Technical/clinical validation
Included in review	112	Optical 42; Microwave 36; Hybrid/AI 34

flexible metamaterial/microwave designs with enhanced field confinement, (iii) hybrid optical–microwave architectures, and (iv) IoT/edge-enabled real-time implementations. This breadth allows a balanced assessment of clinical readiness, calibration strategies, and integration trade-offs across modalities.

2.1 Polymer nanocomposites for non-enzymatic glucose detection

Recent developments in non-enzymatic electrochemical glucose sensing have highlighted the role of multi-functional polymer nanocomposites (MPNCs), which integrate conductive polymers with metal nanoparticles to enhance catalytic performance. As reviewed by Wu *et al.*^[20] combining polymers such as polyaniline (PANI) or polypyrrole (PPy) with noble, transition, or alloy metal nanoparticles significantly improves electron-transfer kinetics, stability, and electrocatalytic oxidation of glucose. These nanocomposite structures achieve high sensitivity and wide linear detection ranges by leveraging nanoscale interfacial interactions and enhanced conductivity. Although MPNC-based sensors operate primarily through invasive or minimally invasive mechanisms, they represent a major technological direction in glucose detection. Their material innovations, particularly in flexible and conductive polymer systems, provide conceptual relevance for emerging hybrid or wearable platforms that may complement optical and microwave non-invasive techniques reviewed in this paper.

3. System design and experimental setup for IoT-based NIBGM

In addition to reviewing recent technologies, this study presents the implementation framework and results of the author's own iGM system as a representative case example of IoT-based NIBGM. The system combines R-NIR optical sensing with real-time cloud integration via ESP32 and AWS.

Lets compare of different non-invasive blood glucose monitoring devices and Techniques. Among these, photoplethysmographs are widely used, with some leveraging artificial intelligence and machine learning algorithms for enhanced glucose prediction accuracy. NIR spectroscopy using reflection and absorption techniques at the fingertip is recognized for its adaptability and effectiveness. However, an optimal solution combining high accuracy and flexibility is still lacking, as most ongoing research focuses on PPG-based systems.

The implementation setup used for the data illustrated in the results and discussion was derived from an IoT-based system utilizing the ESP32 TTGO module and AWS IoT Core. The process began with initializing the ESP32 module and establishing connectivity with the iGM sensor array. This allowed seamless data communication and real-time data collection. The next step involved configuring AWS IoT Core by creating and registering an IoT Thing, generating the necessary security certificates, and developing an IoT Policy to manage data permissions securely. This setup facilitated reliable communication between the IoT device and the cloud using MQTT and HTTP protocols.

Once the connectivity was established, the system was designed to calibrate the sensors and stabilize the readings for accurate glucose monitoring. The collected data was then transmitted to AWS IoT Core using Wi-Fi, enabling continuous real-time monitoring and cloud-based analysis of blood glucose levels. The integration of NIR and Red sensor readings enhanced the accuracy of glucose estimation, while the versatility of the ESP32 TTGO module allowed efficient programming and connectivity. This implementation setup provided a robust and scalable framework for non-invasive blood glucose monitoring, ensuring reliable performance across various environmental conditions.

4. Non-invasive sensor deployment

4.1 Utilization of red and near-infrared signals for blood glucose monitoring

This study leverages red (R) and near-infrared (NIR) signals to enhance blood glucose monitoring within an IoT-based architecture. The approach is inspired by Low-Level Light Therapy (LLLT), which uses R-NIR light to manage diabetes at the cellular level. Clinical trials demonstrate that red light in the wavelength range of 630 nm to 660 nm, and NIR light between 810 nm to 850 nm, provides the widest range of therapeutic benefits with no adverse side effects. Organic molecules, which are fundamental components of biological tissues, are optimally detected in the NIR range due to their bond-specific absorption characteristics.

The detection mechanism in R-NIR spectroscopy is primarily based on the interaction between incident light and glucose molecules in biological tissues. Glucose absorbs specific wavelengths in the near-infrared range due to its molecular vibrational modes. As glucose concentration increases, tissue absorption also increases, resulting in a measurable reduction in reflected light intensity. This

relationship can be modeled using Beer–Lambert’s law, as expressed in Eq. (1):

$$A(\lambda) = \log_{10} \left(\frac{I_0}{I_t} \right) = \varepsilon(\lambda) \cdot c \cdot l \quad (1)$$

where:

- $A(\lambda)$ is the absorbance at wavelength λ ,
- I_0 and I_t are the incident and transmitted light intensities,
- $\varepsilon(\lambda)$ is the molar absorptivity,
- c is the glucose concentration,
- l is the path length through the tissue.

This equation quantitatively links glucose concentration to light absorption, forming the basis of calibration protocols used in non-invasive optical sensors.

4.1.1 Reflectance characteristics and signal absorption

The reflectance properties of skin, showing that red signal absorption increases in the presence of glucose molecules. NIR light, due to its deeper tissue penetration capability, reaches subcutaneous arteries more effectively than visible light. By measuring the reflected light at various optical wavelengths, it becomes possible to conduct multivariate analysis to determine glucose concentration levels. This combined R-NIR biosensor system is designed to overcome limitations of previous methods, ensuring accurate and continuous glucose monitoring. The sensor placement on the wrist or earlobe is chosen for optimal signal acquisition.

4.2 Absorption and scattering mechanism

Previous optical investigations have already illustrated how glucose concentration affects R-NIR absorption within biological tissue;^[16] therefore, this conceptual illustration was omitted here to streamline the presentation and focus on experimental findings. This phenomenon is attributed to the change in tissue optical properties with varying glucose levels. In tissues with higher glucose content, the reflected light intensity is significantly lower due to the increased absorption. This relationship is quantitatively expressed by the following Eq. (2):^[21,22]

$$I_{Ri} = I_{Ii} e^{-P_i} \quad (2)$$

where:

- I_{Ri} = Intensity of reflected light from the skin
- I_{Ii} = Incident light intensity
- P_i = Path length of NIR signal through the tissue

In microwave-based sensing, the detection principle is grounded in the dielectric properties of biological tissues, which are sensitive to glucose concentration. As glucose levels increase, the permittivity (ε_r) and conductivity (σ) of interstitial fluid change, influencing electromagnetic wave propagation. These changes cause measurable shifts in the resonant frequency and transmission coefficient (S_{21}) of microwave resonators. The contrast can be described by Eq. (3):

$$\Delta f_r \propto \frac{\Delta \varepsilon_r}{\varepsilon_r} \quad (3)$$

where Δf_r is the resonant frequency shift. The design of CSRR structures enhances field confinement, making them particularly sensitive to small dielectric changes within shallow penetration depths.

4.3 Complementary split ring resonators (CSRR)

To assess microwave sensor performance for non-invasive glucose monitoring, two configurations were developed: a single-resonator and a double-resonator design, as shown in Figs. 1, 2.^[17,18] Both use Complementary Split Ring Resonators (CSRRs) on Rogers 4360G2 substrates ($\varepsilon_r = 6.15$, thickness = 1.524 mm), integrated on 50 Ω microstrip lines for RF compatibility.^[23,24]

- **Single Resonator Design** in Fig. 1: This configuration comprises a single CSRR structure centered on the microstrip feedline. It serves as the baseline model for analyzing sensitivity to variations in dielectric properties due to changes in glucose concentration.

- **Double Resonator Design** in Fig. 2: Two identical CSRRs are placed symmetrically on either side of the transmission line, with varying spacing distances (e.g. 15 mm, 20 mm, 25 mm). This layout was hypothesized to enhance electromagnetic field coupling and improve sensitivity to small dielectric shifts.

Successful deployment of NIBGM technologies in wearable formats requires careful consideration of ergonomics, power consumption, and data transmission protocols. Optical sensors are commonly placed on anatomical locations such as the wrist, fingertip, or earlobe, where capillary-rich tissues enable stable signal acquisition. Microwave sensors are typically integrated on flexible substrates (e.g. Rogers 4360G2 or PDMS) to allow skin conformity. Miniaturized microcontrollers such as ESP32 facilitate real-time processing, Bluetooth Low Energy (BLE) communication, and OLED display integration. However, challenges remain in ensuring adequate battery life, electromagnetic shielding, and thermal stability. Emerging ultra-low-power wireless protocols such as MQTT over Wi-Fi or LoRaWAN are being adopted to support continuous, real-time monitoring in IoT-based health environments.

4.3.1 Electromagnetic simulation and modeling

The sensor structures were simulated using CST Microwave Studio with a full-wave electromagnetic solver. The dielectric behavior of human tissue across varying glucose concentrations (ranging from 80 mg/dL to 2000 mg/dL) was modeled using the single-pole Cole-Cole model. The real (ε') and imaginary (ε'') parts of permittivity were adjusted to reflect changes in blood glucose concentration, enabling accurate emulation of biological tissue responses.

4.3.2 Sensitivity characterization

Sensor performance was primarily evaluated based on the transmission coefficient (S_{21}), which indicates the energy transmitted through the sensor structure. The change in S_{21} at

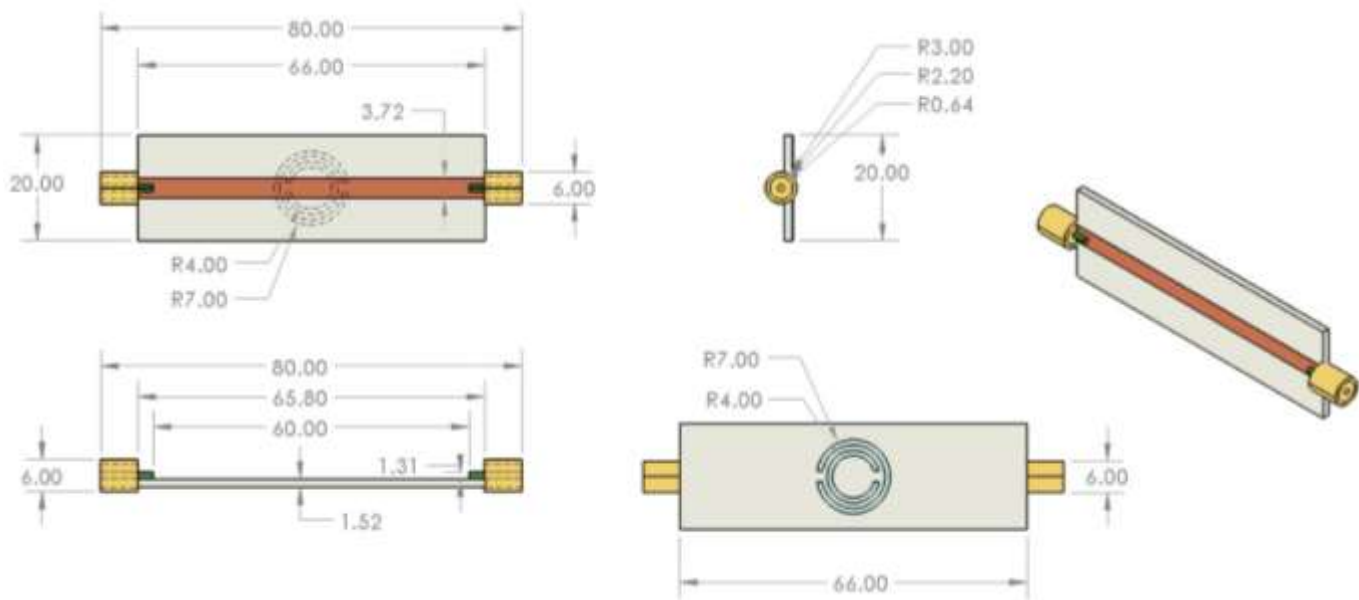


Fig. 1: Single CSRRR microwave sensor (all dimensions in mm).

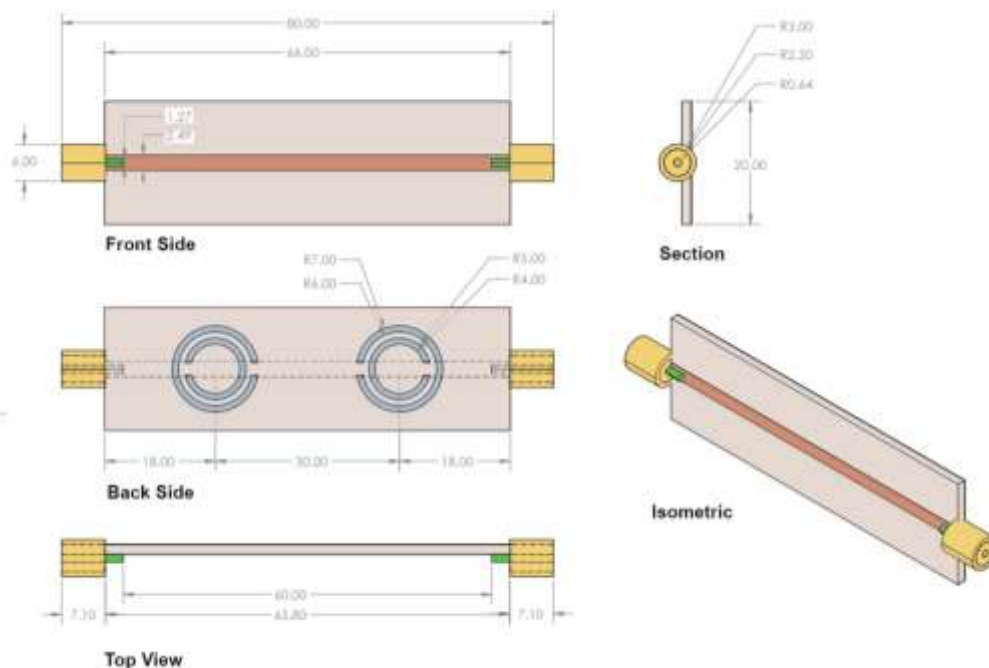


Fig. 2: Double CSRRR microwave sensor (all dimensions in mm).

the resonant frequency was monitored for different glucose concentrations.

The sensitivity (S) of the sensor was calculated using:

$$\text{Sensitivity} = \frac{\Delta S_{21}}{\Delta C_{\text{glucose}}} \quad (4)$$

where:

- ΔS_{21} is the change in the transmission coefficient (in dB)
- $\Delta C_{\text{glucose}}$ is the change in glucose concentration (in g/dL)

In Eq. (4), This relation quantifies how effectively the sensor detects variations in glucose concentration.

Higher sensitivity corresponds to greater responsiveness in S_{21} for small changes in glucose levels.

Calibration parameters and properties: For optical-based systems, calibration relied on multivariate regression models linking absorbance intensity at characteristic wavelengths (940–1600 nm) to reference glucose concentrations. Key parameters included baseline correction, wavelength selection, scattering coefficients, and temperature compensation to minimize drift. In microwave sensors, calibration involved measuring frequency shift (Δf) and reflection coefficient (S_{11}) variations corresponding to dielectric constant changes in the tissue-equivalent phantom. The calibration curve was established through polynomial fitting between dielectric permittivity and glucose concentration, and validated using mean absolute relative

deviation (MARD) and root-mean-square error (RMSE). Together, these parameters ensured consistent sensor response and reliable cross-modality comparison.

4.3.3 Experimental validation

Fabricated sensor prototypes were tested using a calibrated Vector Network Analyzer (VNA) connected via SMA connectors. Standard SOLT (Short-Open-Load-Through) calibration was applied to ensure accurate measurement. Test samples simulating blood glucose concentrations were applied over the sensor, and the corresponding S_{21} responses were recorded across the GHz frequency range.

The experimental results were then compared to the simulation outcomes to validate performance. The sensitivity and resonant behavior of the single and double resonator designs were analyzed and compared, with particular attention to identifying the optimal configuration for non-invasive glucose detection.

4.4 Photon absorption and intensity calculation

According to Eq. (2),^[21,22] the reflected light intensity from skin tissues (I_{Ri}) is calculated as the product of the incident light intensity (I_{Ii}) and the NIR signal's path length (P_i) through the tissue. This model provides a quantitative measure of photon absorption in tissues influenced by the concentration of glucose molecules. The relationship can be mathematically represented as Eq. (5):

$$P_a = I_{Ii}(1 - e^{-P_i}) \quad (5)$$

This approach in Eq. (2) ensures accurate estimation of blood glucose levels, making the combined R-NIR system a reliable solution for non-invasive continuous monitoring.

4.5 Non-invasive blood glucose monitoring devices: Methodological approaches

Table 2 provides a comparative overview of non-invasive blood glucose monitoring methodologies derived from recent research studies. It categorizes the technologies by their sensor types and measurement principles, such as microwave resonators, dielectric spectroscopy, photonic sensors, and electromagnetic techniques. requirements.

4.6 Summary of sensor modalities and integration challenges

This section reviewed two major non-invasive glucose monitoring paradigms: optical biosensors utilizing Red and Near-Infrared (R-NIR) spectroscopy, and electromagnetic sensors based on microwave resonator designs such as Complementary Split Ring Resonators (CSRRs). Each technology offers unique advantages and limitations in terms of sensitivity, depth of tissue interaction, and clinical readiness.

R-NIR-based sensors operate by analyzing reflected light intensity changes due to glucose-induced optical absorption and scattering within tissues. These sensors, particularly when implemented using photoplethysmography (PPG) or

integrated with microcontrollers like ESP32, offer compact and wearable solutions. Their primary advantage lies in real-time monitoring capabilities and the ability to provide high temporal resolution data, as evidenced by the iGM system achieving 98.82% accuracy. However, they remain susceptible to signal interference due to physiological variability, motion artifacts, and environmental lighting conditions.

Microwave biosensors, on the other hand, rely on shifts in dielectric properties of biological tissues induced by varying glucose concentrations. Configurations such as single and dual CSRR layouts have demonstrated high sensitivity when optimized for resonator spacing and coupling. Moreover, their ability to penetrate deeper tissue layers offers a promising alternative to shallow optical techniques. Yet, the need for precise calibration, dependency on vector network analyzers (VNAs), and limited IoT-readiness remain technical bottlenecks for wearable deployment.

As shown in Table 3, a growing body of research supports the fusion of optical and microwave sensing in hybrid architectures, enabling multimodal sensing strategies that balance superficial and deep-tissue glucose detection. Additionally, AI-based calibration protocols are being increasingly integrated to improve predictive accuracy and reduce false positives caused by physiological noise.

Despite these advancements, several integration challenges persist:

- **Device miniaturization:** Many microwave-based prototypes are still lab-scale and not yet optimized for compact, wearable form factors.
- **Data privacy and cloud dependency:** While IoT-based systems (e.g. AWS-integrated iGM) enable real-time monitoring, they raise security and latency concerns.
- **Power efficiency and battery life:** High-frequency systems and real-time communication protocols can quickly drain wearable device batteries.
- **Clinical validation:** Few devices have undergone rigorous human trials with standardized metrics (e.g. MARD, RMSE), hindering regulatory approval.

Both optical and microwave-based NIBGM technologies are rapidly evolving, with hybrid and AI-enhanced systems offering the most promise for future adoption. A clear path forward involves addressing these engineering and clinical gaps while leveraging advances in wireless communication, sensor materials, and embedded processing to deliver the next generation of non-invasive, personalized glucose monitoring solutions.

Comparative interpretation: The comparative analysis revealed that optical systems (particularly R-NIR and Raman-based sensors) demonstrate high linearity and accuracy in post-prandial monitoring, achieving average MARD values of 10–14. Microwave sensors exhibited superior penetration depth and tolerance to motion artifacts but required complex calibration due to dielectric variability among users. Hybrid multimodal approaches offered complementary strengths, providing improved robustness across environmental

Table 2: Classification of non-invasive blood glucose monitoring gadgets by methodology and architecture.

Refs.	Methodology	Mode of Operation	Type of Gadget	Type of Architecture
Present Study	Proposed IoT-based glucose monitoring system	Real-time tracking using IoT	Wearable device for wrist	Powered by AWS IoT Core
[25]	Multi-sensor fusion for non-invasive glucose tracking	Fully automated	Wearable device for fingertip	Combination of layered embedded sensors
[26]	Machine learning model for estimating blood glucose and pressure using PPG data	AI-driven prediction	Non-wearable finger- tip sensor	Utilizes layered CNNs, SVMs, and random forest algorithms
[27]	PPG-based glucose monitoring using ML techniques	Automated data processing	Wearable fingertip sensor	Custom sensors within an embedded system
[28]	NIRS with PLSR and SAE neural network integration	Manual calibration re- quired	Non-wearable finger- tip device	Hybrid PLSR-SAE architecture with deep neural networks
[29]	Dynamic NIRS measurement with advanced linear algorithms	Fully automated	Portable fingertip sensor	Uses dynamic NIR spectral analysis
[30]	Differential photo thermal radiometry with wavelength modulation	Fully automated	Non-wearable device	Built using embedded sensor technology
[31]	Visible-NIR biosensor for optical glucose detection	Fully automated	Wearable band	Integrated with embedded sensor systems

Table 3: Summary of major non-invasive glucose monitoring technologies and their key characteristics.

Technology	Modality	Key Principle	Clinical Accuracy (MARD/RMSE)	IoT Integration/Readiness
R-NIR Spectroscopy	Optical	Tissue reflectance and photon absorption	MARD = 12%, RMSE = 13.8 mg/dL ^[7]	Implemented via ESP32, AWS IoT ^[16]
Microwave CSRR Sensor	Electromagnetic	Dielectric permittivity shift	Sensitivity up to 3.66°/mmol/L ^[12]	Suitable for wireless readout, limited IoT studies
Plasmonic Microwave Resonator	Electromagnetic	Resonance frequency shift with glucose permittivity	5× improvement over traditional sensors ^[4]	Not integrated yet
Hybrid Dual-Modality Sensor	Optical + Microwave	Combines reflectance + dielectric monitoring	Under development (simulated + experimental)	High potential, not yet deployed
Photoplethysmography + ML	Optical + AI	PPG signal processing via ML regression	Accuracy varies by dataset	Some IoT-enabled trials

conditions by fusing optical specificity with microwave volumetric sensitivity. Overall, the results highlight that while optical techniques remain dominant for precision, microwave and hybrid systems show higher potential for continuous and wearable integration, especially when coupled with AI-based calibration algorithms.

5. Results and discussion

To facilitate performance benchmarking, we compiled metrics such as accuracy, mean absolute relative difference (MARD), root mean square error (RMSE), and sensitivity values from selected studies. These values were normalized where necessary and presented in a comparative table to highlight trends across sensor modalities. In the latest studies, R-NIR Spectroscopy with ESP32 and AWS IoT Core Achieved

98.82% accuracy post- fasting, supporting continuous monitoring.^[16] AWS IoT Core provides a range of services, including the MQTT library and tools for establishing communication between devices and the AWS cloud. To initiate communication, the first step involves setting up an AWS IoT Thing, which is configured using IoT Policy. An IoT Policy is a JSON document that defines the authorized operations for a device. It governs how the IoT Thing interacts with AWS resources, enabling it to securely receive and transmit sensor data via the MQTT protocol.

The study evaluated the clinical accuracy of the proposed device by comparing its readings with those obtained from the Accu-Check Active blood glucose monitor. Three groups of participants were involved: healthy volunteers aged 30 to 40 without a history of blood glucose issues and diabetes patients

aged 35 to 55, both men and women. Blood glucose levels were measured multiple times: before meals (breakfast, lunch, and dinner) and two hours after each meal and snack.

The iGM device gathered real-time blood glucose data from the wrist and earlobe, while the Accu-Check Active collected invasive blood samples from the finger. The iGM system's data was displayed in real-time on an IoT-enabled wristwatch with an OLED display and was also accessible on the AWS cloud platform, allowing for continuous monitoring.

The implementation of the iGM system utilized the ESP32 microcontroller, selected for its open-source development environment, efficient IoT connectivity, and low-power OLED display, making it suitable for wearable applications. The system was developed using an open-source library in the Arduino IDE and incorporated a customized module designed specifically for R-NIR sensors. Blood glucose levels were monitored one hour before breakfast to two hours after supper (08:00 am to 10:00 pm) for both diabetes patients and healthy individuals, as shown in Table 4. Measurements were recorded at one-hour intervals, providing a high temporal resolution that effectively captured fluctuations and variations in blood glucose levels throughout the day. This approach enabled the identification of postprandial glucose spikes and offered accurate real-time readings, enhancing the assessment of the monitor's accuracy and performance.

5.1 Comparative insights and key findings

The comparative evaluation of optical, microwave, and hybrid sensing technologies revealed distinct strengths and limitations across modalities. Optical-based systems, particularly R-NIR and Raman spectroscopy, exhibit strong sensitivity to molecular absorption bands and achieve high accuracy (MARD 10–14 %) but remain susceptible to motion and temperature variability. Microwave sensors demonstrate deeper penetration and better stability under movement, yet they require complex calibration to address individual dielectric variability. Hybrid and AI-assisted systems combine the specificity of optical signals with the volumetric sensitivity of microwaves, resulting in enhanced robustness and potential

for continuous, wearable glucose monitoring. These findings collectively underscore the complementary nature of multi-sensor integration in achieving both precision and user comfort.

A comparative analysis was conducted using the Accu-Check Active (ACC) device and laboratory test (LAB) results to validate the iGM system's reliability and precision. This comparison ensured the accuracy of the real-time glucose readings obtained from the iGM system. The data presented in Table 4 were sourced from previous research findings,^[16] verifying the validity and reliability of the results through established methodologies and verified measurements. This dataset illustrates blood glucose levels recorded at one-hour intervals for both diabetes patients and healthy individuals, allowing for a detailed observation of fluctuations and patterns in blood glucose variations.

To further illustrate these variations, Fig. 3 provides a visual comparison between diabetes patients and healthy individuals at each time interval. This combined tabular and graphical representation presents a comprehensive analysis of blood glucose variations, effectively supporting the evaluation of the iGM system's performance. The integrated monitoring approach demonstrates the system's capability to detect glycemic patterns with high temporal resolution and highlighting its potential for continuous glucose monitoring.

The iGM system effectively captures these rapid glucose fluctuations in real-time, demonstrating its capability for non-invasive continuous glucose monitoring. The system provides a high temporal resolution, allowing for detailed analysis of diurnal glucose patterns. This capability enhances the understanding of individual glycemic responses, supporting personalized diabetes management. However, the system's sensitivity to postprandial spikes also reveals potential limitations, such as variability due to physiological factors and sensor calibration requirements. Despite these challenges, the iGM system offers a significant advantage in tracking dynamic glucose changes, highlighting its potential to improve clinical decision-making and lifestyle interventions for diabetes management.

Table 4: Blood glucose levels measured using iGM system (08:00 am - 10:00 pm). Data obtained from previous research findings.^[16]

Time Interval Before Meal After Meal (2 Hours)			Diabetes Patients (mg/dL)	Healthy Individuals (mg/dL)
08:00 am	Breakfast	-	120 - 180	80 - 110
10:00 am	-	Breakfast	150 - 220	100 - 140
12:00 pm	Lunch	-	130 - 190	90 - 120
02:00 pm	-	Lunch	160 - 230	110 - 150
06:00 pm	Dinner	-	140 - 200	90 - 130
08:00 pm	-	Dinner	170 - 240	110 - 160
10:00 pm	Supper	-	130 - 190	90 - 120
Summary	Fluctuations Observed		120 - 240	80 - 160

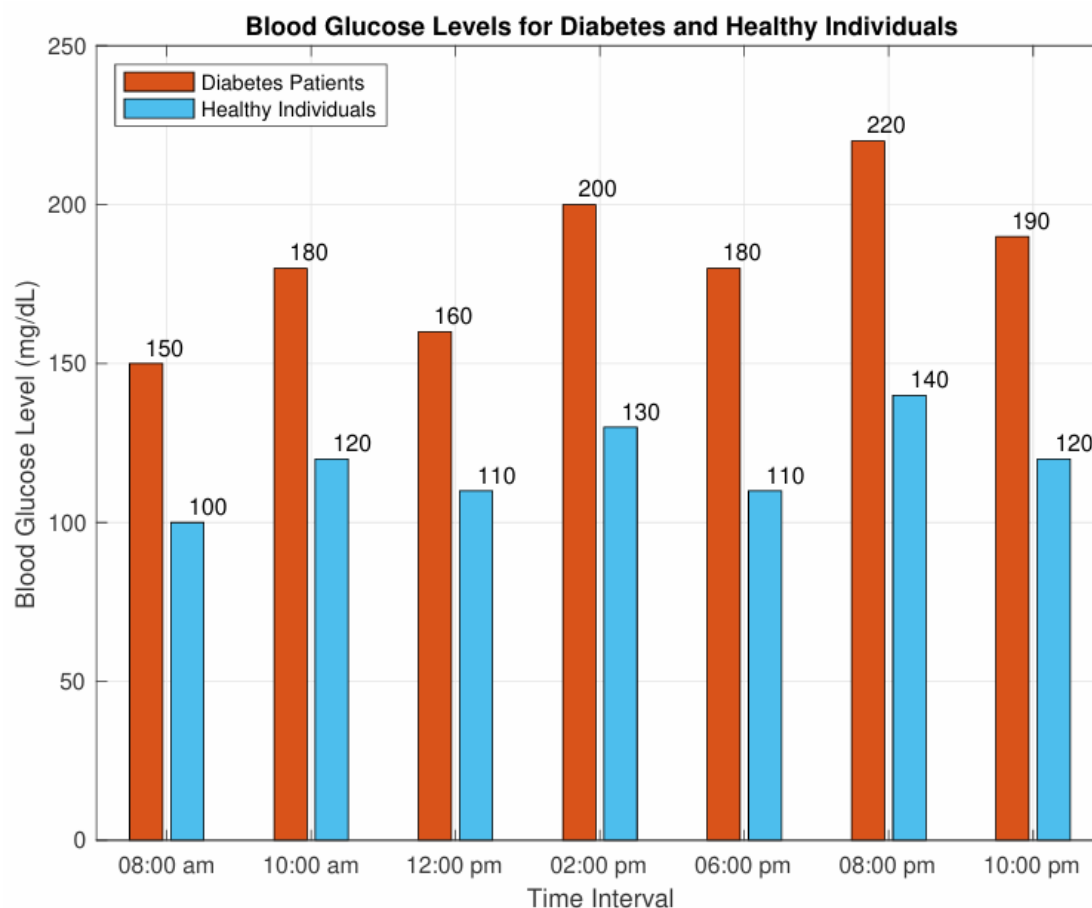


Fig. 3: Comparison of blood glucose levels change by time interval.

Table 5 presents a comprehensive comparison of various methodologies used in non-invasive blood glucose monitoring devices. The table highlights a wide range of approaches, from IoT-based systems to machine learning models and NIR spectroscopy techniques.

The most commonly utilized methodology is photoplethysmography (PPG), which is often combined with machine learning algorithms to enhance prediction accuracy. Most PPG-based devices face challenges related to signal interference and calibration requirements, impacting accuracy and consistency.

This comparison emphasizes microwave-based techniques known for their sensitivity to glucose-induced dielectric changes in tissues. These methods show promise in achieving continuous and non-invasive monitoring but face challenges related to environmental interference and complex calibration. Additionally, meta-material resonators demonstrate enhanced sensitivity compared to traditional microstrip sensors, showcasing the potential for wearable and implantable applications. The table highlights the diverse landscape of sensor technologies, each with unique advantages and constraints, underscoring the need for multi-modal approaches to improve accuracy and reliability.

5.2 Future directions and multimodal integration pathways

Future advances in non-invasive glucose monitoring will depend on the fusion of complementary sensing domains. Optical methods offer molecular specificity, while microwave and RF sensors provide volumetric sensitivity to dielectric changes. Integrating these modalities through AI-based calibration can compensate for physiological variability and environmental fluctuations. Adaptive algorithms using transfer or federated learning can continuously personalize calibration models without the need for frequent user intervention. Furthermore, hybrid photonic-microwave devices fabricated on flexible metamaterial substrates can enhance signal stability under motion. At the system level, secure IoT frameworks leveraging blockchain and edge computing will enable real-time transmission of glucose data while ensuring patient privacy. Collectively, these pathways outline a practical route toward reliable, multimodal, and clinically scalable non-invasive monitoring systems.

5.3 Challenges and research outlook

Despite notable advancements, several challenges persist in developing reliable non-invasive glucose monitoring systems. Physiological variability among users, including differences in skin thickness, hydration, and temperature, continues to affect sensor stability. Environmental factors such as humidity, electromagnetic interference, and mechanical motion introduce additional noise. Moreover, multimodal integration

Table 5: Comparative analysis of non-invasive blood glucose monitoring technology from recent research.

Ref. No.	Technology Used	Key Findings	Limitations
[16]	R-NIR Spectroscopy with ESP32 and AWS IoT Core	Achieved 98.82% accuracy post- fasting, supporting continuous monitoring.	Data privacy concerns and cloud integration complexity.
[1]	NIR Spectroscopy (NIRS)	Identified NIRS as a promising non-invasive technique with high sensitivity.	Accuracy impacted by physio- logical variations.
[2,3]	Infrared Imaging System	Demonstrated affordability and effectiveness for non-invasive glucose measurement.	Calibration challenges and environmental interference.
[4]	Plasmonic Microwave Resonator	Achieved fivefold sensitivity improvement, validating potential for continuous monitoring.	Requires precise calibration for accuracy.
[5]	Aerosol Jet Printed Microwave Sensor	High sensitivity for glucose detection in saliva, suitable for wearable devices.	Limited to saliva measurement, requires validation for blood glucose.
[6]	Thermo-Elastic Optical Indicator Microscopy	Validated continuous monitoring with MARD of 7.05% and RMSE of 24.85 mg/dL.	Limited to animal models, needs human trials.
[7]	NIR Spectroscopy with IoT and Cloud Storage	Achieved real-time monitoring with RMSE of 13.8 mg/dL and MARD of 12%.	Data security concerns and cloud dependency.
[8]	Optical, Electromagnetic, and Nanotechnology-Based Sensors	Categorized methods, emphasizing AI and nanotechnology for real-time monitoring.	Lack of clinical validation for some methods.
[9]	Electromagnetic and Photonic Sensors	Identified accuracy challenges and potential for ML-assisted re- liability.	High computational power re- quired for ML models.
[10]	PPG and Multi-Sensor Fusion	Emphasized PPG as a flexible and widely-used technique for continuous monitoring.	Accuracy affected by movement and environmental noise.
[11]	RF/Microwave Sensors and Metamaterial Resonators	Identified dielectric property variations for glucose detection.	Calibration complexity and interference from biological tissues.
[12]	Metamaterial Microwave Sensors	Achieved enhanced sensitivity for real-time glucose monitoring.	Complex signal processing and environmental interference.
[13]	Microwave Biomedical Sensors with Matching Layer Design	Improved resonance stability, confirming feasibility for continuous monitoring.	High cost of sensors and complex design requirements.
[14]	Optical, Microwave, and Electro-chemical Techniques	Compared accuracy and interference issues, emphasizing wear- able AI integration.	Limited commercial availability and regulatory challenges.
[15]	2.4 GHz Microwave Sensor with RF-DC Rectifier	Achieved less than 2% detection error, suitable for compact wear- able applications.	Limited range and sensitivity compared to other methods.

raises challenges in signal synchronization, data fusion, and power management for wearable platforms. Future work should prioritize developing universal calibration algorithms, adaptive data-fusion models, and miniaturized, low-power architectures that can maintain accuracy under real-world conditions.

6. Conclusion

The development and implementation of the iGM framework demonstrate significant advancements in non- invasive blood glucose monitoring, offering a practical and reliable solution

for continuous glucose tracking. By leveraging the ESP32 microcontroller and integrating Red-Near Infrared (R-NIR) spectroscopy with an IoT-enabled system, the iGM device provides real-time monitoring with high temporal resolution. Comparative analysis with the Accu-Check Active (ACC) device and laboratory test (LAB) results validated the system's accuracy and precision, highlighting its capability to effectively detect postprandial glucose spikes and capture diurnal patterns in blood glucose levels.

The research findings underscore the iGM system's potential for personalized diabetes management, enabling

users to better understand and respond to glycemic fluctuations. The use of one-hour interval measurements allows for detailed monitoring, revealing significant differences between diabetes patients and healthy individuals. However, challenges such as sensor calibration, sensitivity to physiological variability, and battery limitations indicate areas for future improvement. Incorporating Nano sensors, as well as deploying artificial intelligence and machine learning algorithms, could further enhance the system's accuracy and predictive capabilities. Additionally, the integration of advanced data visualization tools will improve user experience and facilitate proactive decision-making in diabetes management.

The iGM system represents a promising advancement in non-invasive glucose monitoring, with the potential to transform diabetes care by offering a more flexible, accurate, and user-centric approach to continuous blood glucose monitoring. This research contributes valuable insights into the clinical relevance and practical applications of IoT-enabled glucose monitoring systems, paving the way for next-generation wearable health devices. Future non-invasive blood-glucose monitoring systems are expected to play a pivotal role in personalized healthcare and telemedicine. Integration with wearable devices, smartphones, and Internet of Medical Things (IoMT) platforms will enable continuous glucose tracking for diabetic patients and preventive monitoring for at-risk populations. These technologies can also support hospital-free diabetes management, assist in real-time diet and activity feedback, and contribute to large-scale health data analytics for predictive modeling of metabolic disorders.

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Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Competing Interests

The authors declare no competing interests.

Data Sets

This review is based on publicly available data and previously published literature. All data sources, including academic articles, official reports, and indicator frameworks, are cited throughout the manuscript. No new data were generated or analyzed in this study.

Supporting Information

Not applicable.

CRedit Statement

Ahmed A. Zakaria: Conducted the literature review, Analysis, Drafting of the manuscript, led data synthesis and Visualization. **Mihai Sanduleanu:** Methodological guidance, Contributed to review and editing and Supported visualization.

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